

Instruments et benchmarks

Compte rendu des TD01 et TD02

Instrument : AAPL – Apple Inc. Benchmark : SP500 – S&P 500

7 octobre 2026

Ce document rassemble les commandes exécutées, les résultats obtenus et les réponses aux questions des deux premiers TD du projet MarketPulse : la prise en main du dépôt en ligne de commande (TD01), puis la lecture des fichiers CSV et JSON en Python (TD02).

TD01 Bootstrap Linux : environnement, exploration du dépôt, lecture des données

TD02 Python, CSV et JSON : dictionnaires, listes, filtrage, fonctions réutilisables

TD01

Bootstrap Linux

Prise en main du dépôt et des données en ligne de commande

1 Vérifier l'environnement

COMMANDE

```
$ python --version
```

RÉSULTAT

```
Python 3.14.6
```

COMMANDE

```
$ git --version
```

RÉSULTAT

```
git version 2.50.1 (Apple Git-155)
```

2 Explorer la racine du dépôt

COMMANDE

```
$ ls -la
```

RÉSULTAT

```
[ls -la
config          evidence      README.md      TEAM_TEMPLATE.md
CONTRIBUTING.md labs          requirements.txt
data            readiness     src
total 56
drwxr-xr-x  14 THOMAS  staff   448  7 oct.  17:48 .
drwxr-xr-x   4 THOMAS  staff   128  7 oct.  17:49 ..
drwxr-xr-x  12 THOMAS  staff   384  7 oct.  18:14 .git
-rw-r--r--   1 THOMAS  staff   178  7 oct.  17:48 .gitignore
drwxr-xr-x   3 THOMAS  staff    96  7 oct.  17:48 config
-rw-r--r--   1 THOMAS  staff  2943  7 oct.  17:48 CONTRIBUTING.md
drwxr-xr-x   3 THOMAS  staff    96  7 oct.  17:48 data
drwxr-xr-x   3 THOMAS  staff    96  7 oct.  17:48 evidence
drwxr-xr-x   6 THOMAS  staff   192  7 oct.  17:48 labs
drwxr-xr-x   3 THOMAS  staff    96  7 oct.  17:48 readiness
-rw-r--r--   1 THOMAS  staff 10270  7 oct.  17:48 README.md
-rw-r--r--   1 THOMAS  staff   133  7 oct.  17:48 requirements.txt
drwxr-xr-x   3 THOMAS  staff    96  7 oct.  17:48 src
-rw-r--r--   1 THOMAS  staff  1446  7 oct.  17:48 TEAM_TEMPLATE.md
```

3 Naviguer vers les données

COMMANDES

```
$ cd data
$ pwd
$ ls

$ cd sample
$ pwd
$ ls
```

RÉSULTAT

```
bloomberg_reference_expected.json  instruments.json
bloomberg_reference_sample.json    prices.csv
```

4 Lire les métadonnées des instruments

COMMANDE

```
$ cat data/sample/instruments.json
```

RÉSULTAT

```
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

5 Afficher les premières lignes de prix

COMMANDE

```
$ head data/sample/prices.csv
```

RÉSULTAT

```
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

6 Filtrer les prix par ticker

COMMANDES

```
$ grep AAPL data/sample/prices.csv  
$ grep SP500 data/sample/prices.csv
```

RÉSULTAT

```
grep SP500 data/sample/prices.csv  
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000  
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000  
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000  
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000  
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000  
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000  
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000  
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000  
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000  
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000  
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000  
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000  
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000  
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000  
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000  
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000  
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000  
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000  
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000  
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000  
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000  
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0  
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0  
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0  
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0  
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0  
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0  
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0  
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0  
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0  
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0  
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0  
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0  
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0  
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0  
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0  
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0  
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0  
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0  
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0  
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0  
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

7 Compter les observations

COMMANDES

```
$ grep -c AAPL data/sample/prices.csv  
$ grep -c SP500 data/sample/prices.csv
```

RÉSULTAT

```
21  
21
```

Chaque série contient 21 observations.

8 Lancer MarketPulse

Depuis la racine du dépôt :

COMMANDE

```
$ python src/main.py
```

RÉSULTAT

```
=== MarketPulse ===  
  
Instrument  
AAPL - Apple Inc.  
Last price: 266.20 USD  
  
Benchmark  
SP500 - S&P 500  
Last level: 6742.00  
  
Period: 1 month  
Interval: Daily  
  
Observations  
AAPL: 21  
SP500: 21
```

9 Questions

RÉPONSES

Quel est l'instrument ? Apple Inc., de ticker AAPL, coté en dollars (USD) sur le NASDAQ.

Quel est le benchmark ? Le S&P 500, noté SP500, exprimé en dollars.

Où sont stockés leurs noms et tickers ? Dans le fichier data/sample/instruments.json.

Exercices optionnels

Why is having two aligned series useful for future comparison work?

RÉPONSE

Aligned series give the instrument and benchmark prices for the same dates, so we compare like with like. This makes later calculations such as returns, correlation or outperformance reliable.

COMMANDES

```
$ ls -lh
$ head -n 5 data/sample/prices.csv
$ tail -n 5 data/sample/prices.csv
```

RÉSULTAT

```
total 48
drwxr-xr-x  3 THOMAS  staff   96B  7 oct.  17:48 config
-rw-r--r--  1 THOMAS  staff  2,9K  7 oct.  17:48 CONTRIBUTING.md
drwxr-xr-x  3 THOMAS  staff   96B  7 oct.  17:48 data
drwxr-xr-x  3 THOMAS  staff   96B  7 oct.  17:48 evidence
drwxr-xr-x  6 THOMAS  staff  192B  7 oct.  17:48 labs
drwxr-xr-x  3 THOMAS  staff   96B  7 oct.  17:48 readiness
-rw-r--r--  1 THOMAS  staff  10K  7 oct.  17:48 README.md
-rw-r--r--  1 THOMAS  staff  133B  7 oct.  17:48 requirements.txt
drwxr-xr-x  3 THOMAS  staff   96B  7 oct.  17:48 src
-rw-r--r--  1 THOMAS  staff  1,4K  7 oct.  17:48 TEAM_TEMPLATE.md
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

COMMANDES

```
$ find . -type f
$ grep -R "AAPL" .
```

RÉSULTAT (EXTRAIT)

```
./git/logs/refs/heads/mabranche
./git/logs/refs/heads/main
./git/logs/refs/remotes/origin/HEAD
./git/logs/refs/remotes/origin/mabranche
./git/description
./git/hooks/commit-msg.sample
./git/hooks/pre-rebase.sample
./git/hooks/sendemail-validate.sample
./git/hooks/pre-commit.sample
./git/hooks/applypatch-msg.sample
./git/hooks/fsmonitor-watchman.sample
./git/hooks/pre-receive.sample
./git/hooks/prepare-commit-msg.sample
./git/hooks/post-update.sample
./git/hooks/pre-merge-commit.sample
./git/hooks/pre-applypatch.sample
./git/hooks/pre-push.sample
./git/hooks/update.sample
./git/hooks/push-to-checkout.sample
./git/refs/heads/mabranche
./git/refs/heads/main
./git/refs/remotes/origin/HEAD
./git/refs/remotes/origin/mabranche
./git/index
./git/packed-refs
./data/sample/bloomberg_reference_sample.json
./data/sample/prices.csv
./data/sample/instruments.json
./data/sample/bloomberg_reference_expected.json
./src/main.py
./labs/TD01_BOOTSTRAP_LINUX.md:Instrument : AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:grep AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL: 21
./labs/TD01_BOOTSTRAP_LINUX.md:instrument : AAPL
./labs/TD01_BOOTSTRAP_LINUX.md:grep -c AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:grep -R "AAPL" .
./labs/TD02_PYTHON_CSV_JSON.md:Instrument : AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] separates AAPL and SP500 rows
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 250.00
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 266.20
./labs/TD02_PYTHON_CSV_JSON.md:Do not duplicate one display implementation for AAPL and another for SI
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] 21 observations for AAPL
./labs/TD02_PYTHON_CSV_JSON.md:[ ] AAPL has 21 observations
./config/settings.yml: ticker: AAPL
./TEAM_TEMPLATE.md:- Instrument: AAPL - Apple Inc.
./README.md:Instrument : AAPL - Apple Inc.
./README.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL - Apple Inc.
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL: 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep -c AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:[x] AAPL observation count is 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./data/sample/bloomberg_reference_sample.json: "canonical_ticker": "AAPL",
```

Python, CSV et JSON

Charger, filtrer et résumer les données de marché avec des fonctions réutilisables

1 Découvrir le programme de départ

COMMANDE

```
$ python src/main.py
```

RÉSULTAT

```
=== MarketPulse ===  
  
Instrument  
AAPL - Apple Inc.  
Last price: 266.20 USD  
  
Benchmark  
SP500 - S&P 500  
Last level: 6742.00  
  
Period: 1 month  
Interval: Daily  
  
Observations  
AAPL: 21  
SP500: 21
```

RÉPONSES

Q1. Which function reads JSON?

`load_instruments()` : elle ouvre `instruments.json` et le lit avec `json.load`.

Q2. Which function reads CSV?

`load_prices()` : elle ouvre `prices.csv` et le lit avec `csv.DictReader`.

Q3. Which function selects one ticker?

`filter_prices()` : elle ne garde que les lignes de prix du ticker demandé (AAPL ou SP500).

Q4. Which function coordinates execution?

`main()` : elle appelle les autres fonctions dans l'ordre, puis affiche le résultat.

Q5. Where are the input files stored?

Dans le dossier `data/sample`, dont le chemin est enregistré dans la variable `DATA_DIR`. Il contient `instruments.json` et `prices.csv`.

2 Comprendre le JSON et les dictionnaires

COMMANDE

```
$ cat data/sample/instruments.json
```

RÉSULTAT

```
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

CODE D'INSPECTION TEMPORAIRE

```
print(type(instruments))
print(instruments.keys())

instrument = instruments["instrument"]
benchmark = instruments["benchmark"]

print(instrument["ticker"])
print(instrument["name"])
print(instrument["currency"])

print(benchmark["ticker"])
print(benchmark["name"])
```

RÉSULTAT

```
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
```

`instruments` est un dictionnaire qui contient deux objets métier : `instrument` décrit l'instrument principal et `benchmark` décrit l'indice de comparaison.

3 Comprendre le CSV et les listes de dictionnaires

COMMANDE

```
$ head data/sample/prices.csv
```

RÉSULTAT

```
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

Les colonnes sont : date, ticker, open, high, low, close, volume.

CODE D'INSPECTION TEMPORAIRE

```
prices = load_prices()

print(type(prices))
print(type(prices[0]))
print(prices[0])

print(prices[0]["close"])
print(type(prices[0]["close"]))

close = float(prices[0]["close"])
print(close)
print(type(close))
```

RÉSULTAT

```
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low':
  '248.00', 'close': '250.00', 'volume': '38000000'}
250.00
<class 'str'>
250.0
<class 'float'>
```

prices est une liste dont chaque élément est un dictionnaire. Les valeurs lues dans un CSV sont du texte (str) : il faut les convertir avec float() avant tout calcul.

4 Filtrer l'instrument et le benchmark

CODE

```
instrument_prices = filter_prices(prices, instrument["ticker"])
benchmark_prices = filter_prices(prices, benchmark["ticker"])

print(len(instrument_prices))
print(len(benchmark_prices))
```

RÉSULTAT

```
Observations
AAPL: 21
SP500: 21
```

Le filtre renvoie bien 21 observations pour chaque série.

5 Premier et dernier cours de clôture

FONCTIONS CRÉÉES

```
def get_first_close(prices):
    first_row = prices[0]
    return float(first_row["close"])

def get_last_close(prices):
    last_row = prices[-1]
    return float(last_row["close"])
```

RÉSULTAT

```
250.0
6600.0
266.2
6742.0
```

Premier cours : 250.00 pour AAPL et 6600.00 pour SP500. Dernier cours : 266.20 pour AAPL et 6742.00 pour SP500. Ce sont les valeurs attendues.

6 Un résumé de marché réutilisable

Une seule fonction, `display_market_summary`, affiche le résumé de l'instrument comme celui du benchmark.

RÉSULTAT

```
Instrument
AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00
```

7 Code final : un `main()` lisible

La fonction `main()` enchaîne six étapes : charger les métadonnées, charger les prix, sélectionner l'instrument et le benchmark, filtrer les deux séries, afficher la configuration, puis afficher les deux résumés.

SRC/MAIN.PY

```
from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")

# These starter values mirror config/settings.yml.
# settings.yml is a human-readable configuration contract in the CORE.
# Parsing YAML is optional and is not required by the 18-hour lab sequence.
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

def get_first_close(prices):
    first_row = prices[0]
    return float(first_row["close"])

def get_last_close(prices):
```

```

last_row = prices[-1]
return float(last_row["close"])

def display_market_summary(asset, prices, show_currency=True):
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

    if show_currency:
        suffix = f" {asset['currency']}"
    else:
        suffix = ""

    print(f"{asset['ticker']} - {asset['name']}")
    print(f"Observations : {len(prices)}")
    print(f"First close   : {first_close:.2f}{suffix}")
    print(f>Last close    : {last_close:.2f}{suffix}")

def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print("=== MarketPulse ===")
    print()
    print("Market configuration")
    print(f"Period   : {LOOKBACK_LABEL}")
    print(f"Interval : {INTERVAL_LABEL}")
    print()
    print("Instrument")
    display_market_summary(instrument, instrument_prices)
    print()
    print("Benchmark")
    display_market_summary(benchmark, benchmark_prices, show_currency=False)

if __name__ == "__main__":
    main()

```